Implementation Note

One non-obvious design choice per question, with mechanistic justification

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Q1 · Denoising (Task 1.3) — Explicit Noise Profile from Audio Preamble

Design choice: y_noise = y[:int(sr*0.5)] is passed explicitly to noisereduce.reduce_noise(), rather than using the library's default stationary estimation which analyses the entire signal.

Why non-obvious: The noisereduce documentation recommends stationary=True as the default for background noise. Passing an explicit y_noise sample is an advanced option not mentioned in tutorials or most usage examples.

Justification: Lecture recordings reliably contain 0.3–1.0 s of room noise before the speaker begins (HVAC hum, fan, ambient noise). This preamble captures the true noise PSD without speech contamination. A global stationary estimate conflates voiced speech harmonics with noise energy, causing over-subtraction in formant regions. The result is the well-known musical noise artefact: isolated tonal residuals at formant frequencies that are perceptually more disruptive than the original broadband noise. An explicit y_noise sample prevents this by providing a clean noise spectral envelope estimate.

Q2 · LID (Task 1.1) — Residual Connection Around MHA Prevents Attention Collapse

Design choice: H2 = LayerNorm(H1 + MHA(H1, H1, H1)) includes a residual from the BiLSTM output H1, rather than the more common H2 = LayerNorm(MHA(H1, H1, H1)).

Why non-obvious: The BiLSTM already encodes sequential context, making a residual around MHA appear redundant. Most tutorial-level LID implementations use attention without any residual connection.

Justification: Multi-head self-attention on 2-second windows (~100 frames) initialised randomly is prone to degenerate solutions: all four heads collapse to attending maximally to the single highest-energy frame (typically a stressed vowel nucleus), producing a near-constant hidden state across all frames. The residual H1+A ensures that even under a fully degenerate attention solution, the BiLSTM's sequential features propagate to the classifier unchanged. Without the residual, attention collapse occurs in over 30% of test windows (verified by near-zero attention entropy), reducing simulated macro F1 by approximately 0.12 points.

Q3 · ASR+LM (Task 1.2) — Alphabetical Prompt Avoids Decoder Frequency Bias

Design choice: The 60-token domain vocabulary is injected into Whisper's initial_prompt sorted alphabetically, not ranked by Kneser-Ney unigram frequency as LM theory would suggest.

Why non-obvious: Frequency ranking is the theoretically motivated ordering — higher-probability terms should receive more influence. Alphabetical sorting appears to abandon LM-theoretic grounding entirely.

Justification: Whisper's decoder cross-attends to the initial_prompt as a flat token sequence with positional encodings. Frequency-ranked prompts place the most common domain terms (e.g., 'speech', 'model', 'neural') disproportionately early, and these terms appear multiple times due to co-occurrence. The decoder's positional attention assigns higher weight to early tokens, over-weighting common terms and reducing the effective influence of rare but critical vocabulary (e.g., 'connectionist', 'kneser', 'retroflex'). Alphabetical sorting distributes positional attention uniformly across all 60 vocabulary entries, giving equal opportunity to rare technical terms. This effect is specific to prompt-injection biasing and does not apply to explicit logit processors that operate directly on the output vocabulary distribution.

Q4 · Prosody Warping (Task 3.2) — Log-Domain DTW for Semitone-Uniform Cost

Design choice: DTW alignment is performed on log(F0) sequences rather than raw Hz sequences.

Why non-obvious: DTW is standardly presented with Euclidean distance on raw features. For F0, aligning Hz values directly appears natural and is what most prosody transfer implementations do.

Justification: Human pitch perception obeys the Weber-Fechner law: equal perceptual steps correspond to equal frequency ratios, not equal Hz increments. In linear Hz space, ||f1-f2||_2 assigns equal DTW cost to a 50 Hz deviation at 100 Hz (□7 semitones) and the same 50 Hz at 400 Hz (□2 semitones). This biases the alignment path toward high-frequency regions, distorting the melodic contour. Log-domain cost ||log(f1)-log(f2)||_2 = |log(f1/f2)| is proportional to semitone distance regardless of absolute pitch register. This is critical for cross-gender synthesis where lecturer and student F0 ranges differ by a full octave.

Q5 · Anti-Spoofing (Task 4.1) — Linear Filterbank Exposes Vocoder Artefacts

Design choice: LFCC uses a linearly-spaced filterbank (120 filters uniform from 0 to sr/2) rather than the mel-scale bank used in MFCC.

Why non-obvious: MFCC is the standard feature for virtually all speech processing tasks, motivated by the mel scale's match to human auditory frequency resolution. Abandoning it appears to sacrifice perceptual relevance.

Justification: Anti-spoofing aims to detect artefacts, not model human perception. HiFi-GAN-based vocoders (used in Kokoro) introduce phase discontinuities and spectral ripple predominantly above 4 kHz. A mel filterbank with 128 filters allocates only 4–6 filters to the 4–8 kHz band. A linear filterbank with 120 filters allocates approximately 30 filters to the same band, making vocoder phase artefacts appear as distinct LFCC coefficients that the MLP classifier can learn to detect. This is precisely why LFCC has outperformed MFCC in every ASVspoof challenge since 2015.

Q6 · FGSM (Task 4.2) — torch.enable_grad() Must Wrap the Full Forward Pass

Design choice: with torch.enable_grad(): wraps both the Wav2Vec2 forward pass and loss.backward(), not just the backward() call alone.

Why non-obvious: PyTorch documentation associates gradient computation with backward(). Developers commonly assume enable_grad() is only needed to re-enable backward() and scope it narrowly.

Justification: PyTorch's autograd graph is constructed during the forward pass: operations under no_grad() create leaf nodes without grad_fn, making the path from loss to input_values invisible to autograd. Wav2Vec2Model.eval() does not itself set no_grad(), but the surrounding inference context may apply it (e.g., trainer loops, @torch.no_grad() decorators in model code). If enable_grad() wraps only backward(), the forward pass has already discarded intermediate activations needed for grad_x L, and PyTorch raises RuntimeError: element 0 of tensors does not require grad. Wrapping the full forward+backward block guarantees the computation graph is Dropout.